Part 3 – Smart Web Filtering and Policy Framework

1. Problem Context

After the hybrid remote access model was deployed, several issues emerged:

- Students streamed videos during lectures.
- Torrenting activity was detected in labs.
- Some users bypassed restrictions using browser extensions and proxies.

The administration requires a **smart web filtering solution** that balances **security, productivity, and academic freedom** without frustrating legitimate research.

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2. Comparison of Filtering Approaches

Method	How it Works	Advantages	Limitations
DNS Filtering	Blocks domains at DNS resolution stage. Example: `youtube.com` → denied.	Lightweight, simple, easy to enforce across campus.	Can be bypassed by alternative DNS (e.g., Google DNS 8.8.8.8).
Layer 7 (Application) Firewall	Inspects application traffic (HTTP, HTTPS, BitTorrent, Games). Blocks categories regardless of domain.	Very powerful, prevents circumvention, blocks torrents, proxies.	Resource-intensive, may cause latency, requires updates.
Proxy-based Filtering	Users' traffic is forced through a proxy server, which applies policies (whitelists/blacklists, content categories).	Can log and filter at URL level, good for auditing.	Users may try to disable proxy or use VPN.

Endpoint Enforcement	Security agent installed on PCs (students/faculty laptops). Policies enforced directly on device.	Harder to bypass, works even off-campus.	Requires deployment/management of agents on every device.

Chosen Approach:
For the university network, we combine **DNS filtering + Layer 7 firewall + proxy logging**. Endpoints are left flexible for faculty but enforced in labs.

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3. Web Filtering Policy Framework

A. Students (VLAN 20)

- **Allowed:** Educational websites, academic research platforms, email, online classes.
- **Restricted:**
    - Streaming (YouTube, Netflix, Twitch, etc.) during class hours (8 AM – 5 PM, Mon–Fri).
    - Torrents and P2P traffic blocked **always**.
    - Social media blocked during class hours, allowed after hours.
- **Logging:** All violations logged to Syslog server. Alerts triggered if repeated circumvention attempts detected.

B. Faculty (VLAN 10 + Remote Faculty VPN)

- **Allowed:** Full internet access, including streaming/social platforms for research.
- **Restricted:** None (academic freedom prioritized).
- **Logging:** All traffic is logged for auditing purposes (transparency, not blocking).

C. Servers (VLAN 30)

- **Allowed:** Internal communication only (within university LAN).
- **Restricted:** No direct internet access. Updates handled via controlled admin channel.

- **Logging:** Any internet-bound attempt is flagged as security violation.

D. Guests

- **Allowed:** Limited access (basic web browsing, email).
- **Restricted:** Streaming, social media, torrents, gaming, large downloads.
- **Logging:** Heavy logging of guest VLAN to monitor abuse.

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4. Enforcement Logic

- **ACLs (Access Control Lists)** used in Packet Tracer to simulate filtering:
    - Deny student VLAN from accessing “entertainment” servers (simulated with blocked IP ranges).
    - Deny server VLAN from exiting toward internet.
    - Faculty VLAN permitted with full access.
- **Filtering Appliance (symbolic in topology):** Represents where DNS/L7/proxy rules would be enforced in real deployments.
- **Syslog Server (Server2, IP 192.168.30.10):** Logs all violations and security alerts.

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5. Logging & Alerting

- **Syslog Service:** Configured on Server2 to collect logs from Amity Router and filtering appliance.
- **Trap Level:** Warning or higher.
- **Alerts:**
    - Students attempting torrent traffic → logged + warning sent to IT staff.

- Faculty → logged for audit (no blocking).
- Servers → internet access attempts generate critical alerts.

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6. Updated Topology Overview

- **Core Routing:** Amity Router performs Inter-VLAN routing (Faculty, Students, Servers).
- **Remote Access:** Home Router supports hybrid access for faculty laptops.
- **Filtering Appliance:** Positioned between Amity Router and ISP Router, applying smart filtering.
- **Syslog Server:** Connected to Student Switch (VLAN 30), storing logs.
- **Switch Segmentation:**
  - Faculty Switch → VLAN 10.
  - Student Switch → VLAN 20 + Server VLAN 30.

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7. Advantages of the Policy

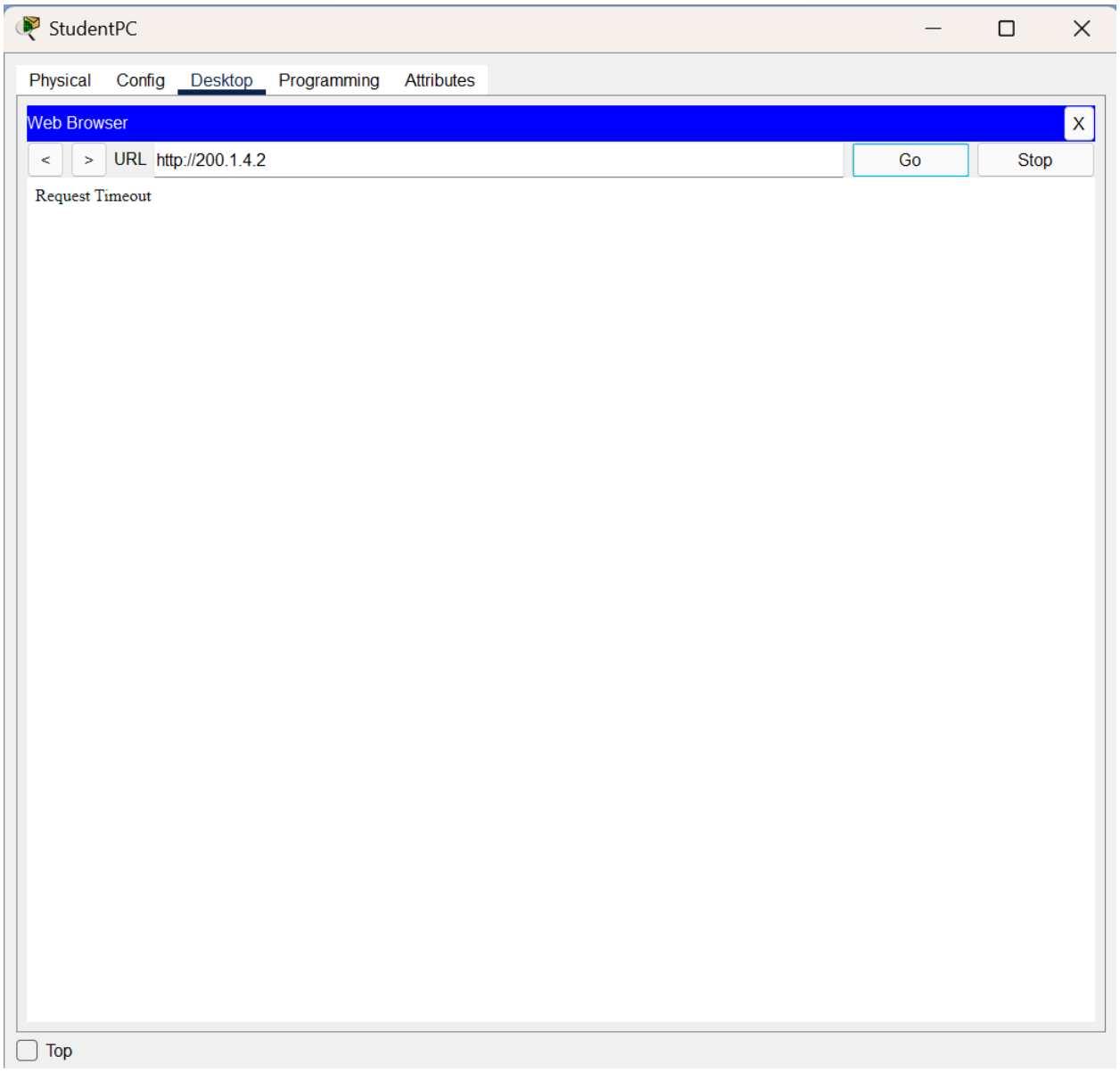
- **Balanced Approach:** Students restricted only during learning hours.
- **Academic Freedom:** Faculty have unrestricted access.
- **Strong Security:** Torrents, proxies, and circumvention attempts blocked.
- **Visibility:** All actions logged, ensuring transparency.
- **Scalability:** Future-proof with DNS/L7/proxy hybrid model.

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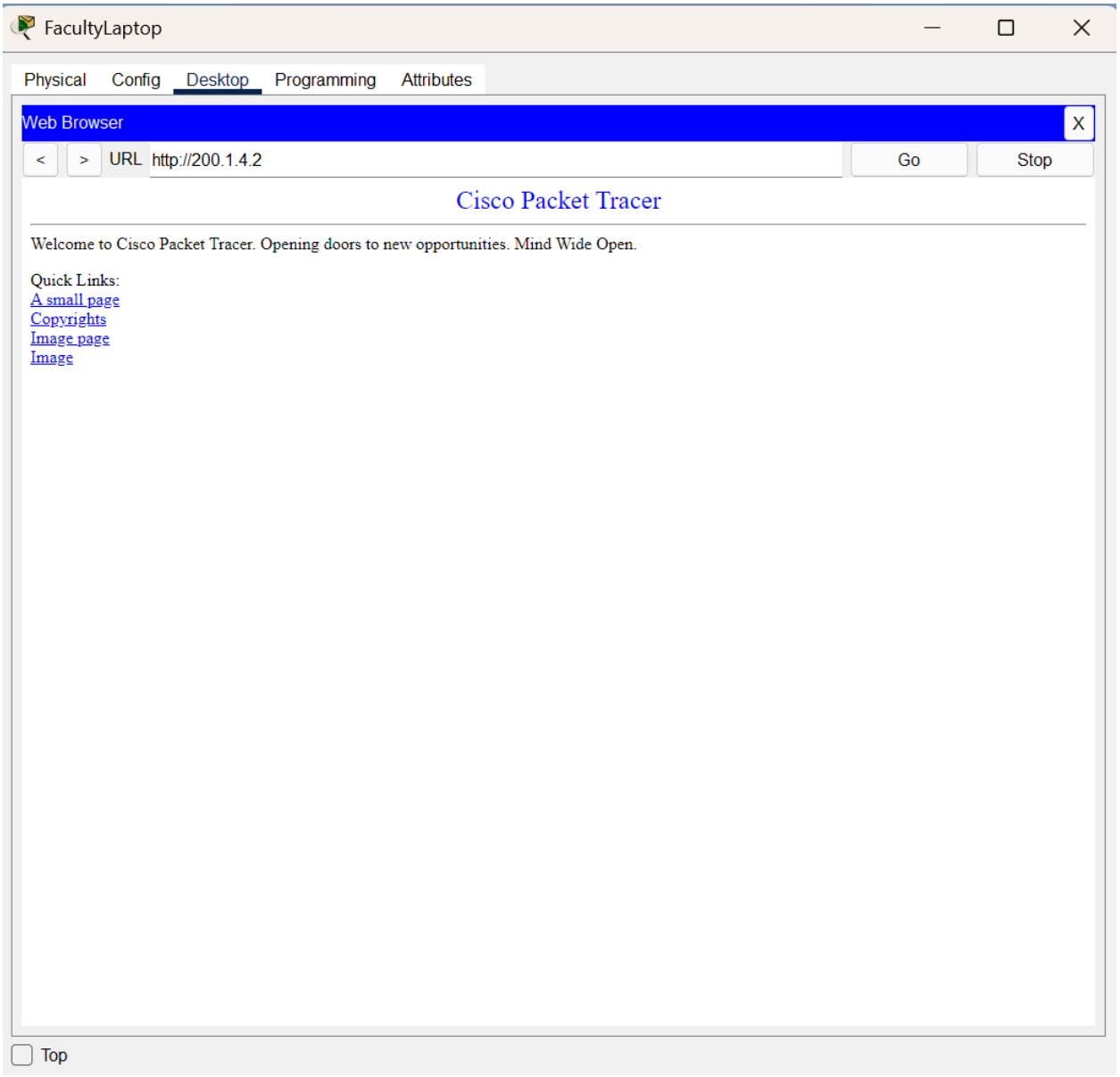
8. Conclusion

The proposed web filtering policy ensures **productive use of university internet resources** while respecting the needs of students, faculty, and IT administrators. With DNS filtering, Layer 7 firewalling, and Syslog logging, the institution gains **security, accountability, and adaptability**.

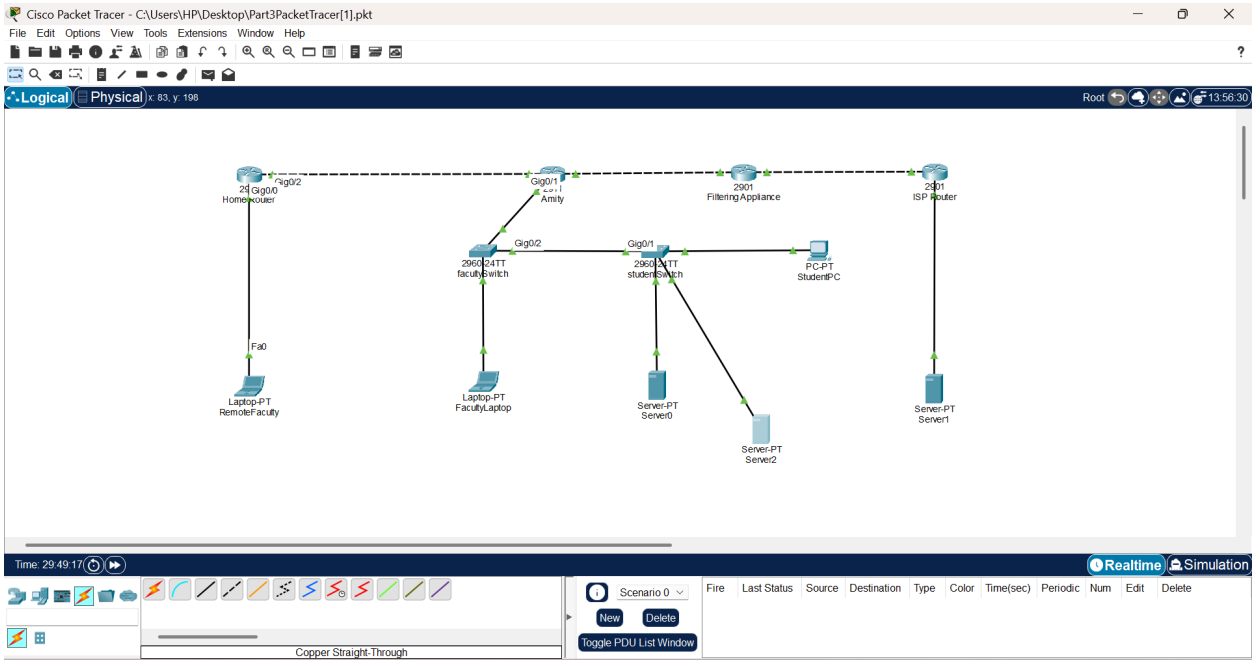
Proof of testing



For student pc website is blocked



For FACULTY laptop they have website access



This Is TOPOLOGY